Chapter 2

A SOFTWARE RELIABILITY MODEL WITH TESTING COVERAGE AND IMPERFECT DEBUGGING

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1. Introduction

Today, computer software is the heart of various safety-critical and civilian systems and software reliability measurement has become a significant factor in quantitatively characterizing quality and determining when to stop testing and release software on the basis of predetermined reliability objectives. Research activities in software reliability engineering have been conducted and several software reliability growth models [Goel and Okumoto (1979), Ohba (1984a, 1984b), Ohba and Yamada (1984), Pham (1993), Pham and Zhang (1997), Pham *et al.* (1999), Tohma *et al.* (1991), Yamada *et al.* (1983), Yamada *et al.* (1986), Yamada *et al.* (1992), Wood (1996)] have been proposed to estimate the number of remaining faults and assess the reliability of software.

NHPP Software reliability models have been successfully used in modeling and analyzing the reliability of the software products. These models consider the debugging process as a counting process, which follows Poisson distribution with a time dependent failure intensity function. Existing software reliability models can be unified into a general form from which different software reliability growth models can be developed [Pham and Zhang (1997)].

Some previous models [Goel and Okumoto (1979), Ohba (1984b), Ohba and Yamada (1984)] were derived based upon perfect debugging which assumes that no new faults are introduced during the debugging process. Others consider imperfect debugging [Yamada *et al.* (1983), Ohba (1984b), Pham (1993), Yamada *et al.* (1986), Yamada *et al.* (1992)] in the sense that new faults can be introduced while the detected faults are removed. The imperfect debugging models assume that the fault content function is nondecreasing as one progresses through the testing phase.

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This indicates that the total number of faults including the initial faults and the introduced ones is increasing in testing time. Recently, a software reliability model based on NHPP that integrates imperfect debugging with the learning phenomenon is developed [Pham *et al.* (1999)].

Another important issue in the software development process is testing coverage. Testing coverage measurement helps software developers to evaluate the reliability of the software during the testing phase and determine how much additional effort is needed to improve the reliability of the software. Testing coverage also can provide customers with a quantitative confidence when they plan to use the software. To our knowledge, testing coverage has not been addressed in the existing imperfect software reliability models.

In this paper, a software reliability model is proposed to integrate testing coverage and imperfect debugging by incorporating fault introduction phenomenon into software detection rate function. In Sec. 2, a new NHPP model addressing fault introduction and fault detection rate is formulated. In Sec. 3, the proposed model is evaluated to illustrate the results by using two sets of data collected from real applications.

1.1. *Notation*

$m(t)$	expected number of software failures by time t, $m(t) = E[N(t)]$
a	number of initial software faults
$d(t)$	imperfect debugging intensity rate
$c(t)$	testing coverage function, i.e., the percentage of the code covered by testing cases by time t
$\lambda(t)$	intensity function or fault detection rate per unit time, $\lambda(t) = d[m(t)]/dt$
$R(x\|t)$	software reliability function for a mission time x given that the last failure occurred at time t
y_k	actual number of failures observed by time t_k

2. Software Reliability Modeling

The general NHPP software reliability growth model is formulated based on the following assumptions:

(1) The occurrence of software failures follows an NHPP.
(2) The software failure intensity rate at any time is proportional to the number of remaining faults in the software at that time.
(3) When a software failure occurs, a debugging effort takes place immediately. This debugging is s-independent at each location of the software failures.
(4) During the debugging process, the effort to remove each fault may not be perfect and therefore new faults may be introduced into the software system with the imperfect debugging intensity rate $d(t)$.

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(5) The imperfect debugging rate is assumed to decrease as testing progresses and becomes negligible towards the end of the testing phase because the experience and knowledge of the testing team increases with the progress of the learning process.
(6) The imperfect fault debugging does not increase the number of initial faults.

2.1. *The Nonhomogeneous Poisson Process*

Software testing process has been widely modeled as an fault counting process. A counting process $\{N(t), t \geq 0\}$ is said to be a nonhomogeneous Poisson process with intensity function $\lambda(t)$, if $N(t)$ follows a Poisson distribution with mean value function $m(t)$, i.e.,

$$\Pr\{N(t) = k\} = \frac{[m(t)]^k}{k!} e^{-m(t)} \qquad k = 0, 1, 2\Lambda \tag{1}$$

The mean value function $m(t)$, which is the expected number of failures experienced up to a certain time t, can be expressed in terms of failure rate of the program, i.e.,

$$m(t) = \int_0^t \lambda(s) ds$$

where $\lambda(s)$ is the failure intensity function.

Software reliability $R(x/t)$ is defined as the probability that a software failure does not occur in $(t, t + x)$, given that the last failure occurred at testing time $t(t \geq 0, x > 0)$. That is,

$$R(x/t) = e^{-[m(t+x)-m(t)]} \tag{2}$$

2.2. *Model Formulation*

In a recent literature on software reliability engineering, imperfect debugging is usually considered by modeling a time-dependent fault content rate function [Ohba (1984a, 1984b), Ohba and Yamada (1984), Pham (1993), Pham and Zhang (1997), Pham *et al.* (1999), Yamada *et al.* (1983), Yamada *et al.* (1986), Yamada *et al.* (1992)]. That is, the total number of faults including the initial faults in the code and the introduced ones increases in testing time. These models require users to select an appropriate fault content rate function. In this paper, we derive a new model which incorporates fault introduction phenomenon into fault detection rate function. By doing so, we take imperfect debugging into the construction of fault detection phenomenon and the number of initial faults can be estimated consistently.

Since fault introduction rate is taken into consideration, a general NHPP model incorporating testing coverage and imperfect debugging can be formulated as follows:

$$\frac{dm(t)}{dt} = \frac{c'(t)}{1 - c(t)}[a - m(t)] - d(t)[a - m(t)] \tag{3}$$

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where a is the number of initial faults in the software code. The function $c'(t)/[1-c(t)]$ represents the fault detection rate function, where $c(t)$ represents the testing coverage function. The function $c(t)$ measure the percentage of the software code covered by testing cases up to any time t. Then, $1-c(t)$ is the percentage of the software code which has not yet been covered by test cases by time t. The derivative of the testing coverage function, $c'(t)$, represents the coverage rate. Therefore, the fault detection rate function can be expressed as $c'(t)/[1-c(t)]$. And $d(t)$ denotes the fault introduction rate which is a decreasing function of time.

Define $g(t) = c'(t)/[1-c(t)]-d(t)$, which represents the imperfect fault detection rate. Equation (1) can be re-written as follows:

$$\frac{dm(t)}{dt} = g(t)[a - m(t)] \tag{4}$$

This model indicates that the failure intensity depends on both the imperfect fault detection rate and the number of remaining faults. Once the imperfect fault detection function, $g(t)$ is determined from Eq. (3) in Pham *et al.* (1999), the mean value function can be expressed as follows:

$$m(t) = e^{-B(t)}\left[m_0 + \int_{t_0}^{t} ae^{B(\tau)}g(\tau)d\tau\right] \tag{5}$$

where $B(t) = \int_{t_0}^{t} g(\tau)d\tau$ and $m(t_0) = m_0$ is the marginal condition of Eq. (2) with t_0 representing the starting time of the debugging process. The reliability function can be derived substitute Eq. (5) into Eq. (3).

2.2.1. *A Software Reliability Model with Testing Coverage and Imperfect Debugging*

In this study, fault introduction phenomenon is considered in the construction of fault detection rate function. The testing coverage function is a nondecreasing function of testing time t (see Fig. 1). The following function can be used to model the testing coverage function

$$c(t) = 1 - (1 + bt)e^{-bt} \tag{6}$$

then

$$\frac{c'(t)}{1-c(t)} = \frac{b^2t}{1+bt} \tag{7}$$

This is an S-shaped curve that captures the learning process of software tester [Yamada *et al.* (1983), Ohba and Yamada (1984)]. In other words, the fault detection rate increases at beginning but with a slower slope; after a certain period, since testers earn knowledge and get familiar with the underlying software, the fault detection rate increases faster; at the end most of the faults have been removed and the fault detection turns flat at the tail.

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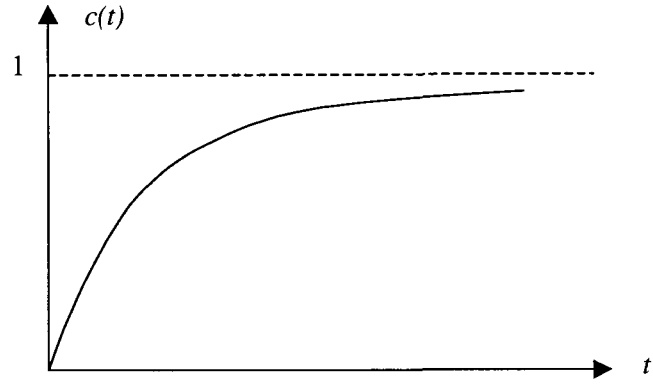


Fig. 1. Testing coverage function $c(t)$.

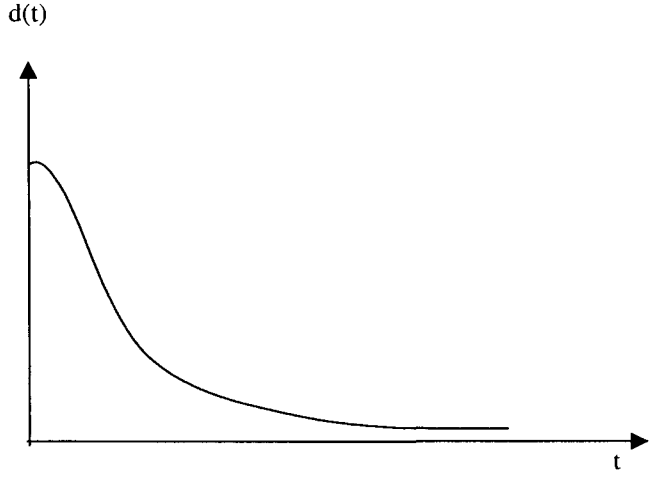


Fig. 2. Fault introduction rate function $d(t)$ versus testing time.

Define the fault introduction rate, $d(t)$, which is a decreasing function of testing time t (see Fig. 2), as follows:

$$d(t) = \frac{d}{1+dt} \tag{8}$$

This function is used to reflect the fact that as one progresses through the testing phase, the fault introduction rate slow down since testers become more experienced and introduce less faults.

The imperfect fault detection rate function is given by:

$$g(t) = \frac{b^2 t}{1+bt} - \frac{d}{1+dt} \tag{9}$$

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Table 1. Summary of the NHPP software reliability models.

Model Name	Model Type	MVF ($m(t)$)	Comments
Goel-Okumoto (G-O) (1979)	Concave	$m(t) = a(1 - e^{-bt})$ $a(t) = a$ $b(t) = b$	Also called exponential model
Delayed S-shaped [Yamada *et al.* (1983)]	S-shaped	$m(t) = a(1 - (1 + bt)e^{-bt})$	Modification of G-O model to make it S-shaped
Inflection S-shaped SRGM [Ohba (1984b)]	Concave	$m(t) = \frac{a(1-e^{-bt})}{1+\beta e^{-bt}}$ $a(t) = a$ $b(t) = \frac{b}{1+\beta e^{-bt}}$	Solves a technical condition with the G-O model. Becomes the same as G-O if $\beta = 0$
Yamada Exponential [Yamada *et al.* (1986)]	Concave	$m(t) = a(1 - e^{-r\alpha(1-e^{(-\beta t)})})$ $a(t) = a$ $b(t) = r\alpha\beta e^{-\beta t}$	Attempt to account for testing effort
Yamada Rayleigh [Yamada *et al.* (1986)]	S-shaped	$m(t) = a(1 - e^{-r\alpha(1-e^{(-\beta t^2/2)})})$ $a(t) = a$ $b(t) = r\alpha\beta e^{-\beta t^2/2}$	Attempt to account for testing effort
Yamada Imperfect Debugging (1) [Yamada *et al.* (1992)]	S-shaped	$m(t) = \frac{ab}{\alpha+b}(e^{\alpha t} - e^{-bt})$ $a(t) = ae^{\alpha t}$ $b(t) = b$	Assume exponential fault content function and constant fault detection rate
Yamada Imperfect Debugging (2) [Yamada *et al.* (1992)]	S-shaped	$m(t) = a[1 - e^{-bt}]\left[1 - \frac{\alpha}{b}\right] + \alpha at$ $a(t) = a(1 + \alpha t)$ $b(t) = b$	Assume constant introduction rate α and the fault detection rate
PNZ [Pham *et al.* (1999)]	S-shaped concave	$m(t) = \frac{a[1-e^{-bt}][1-\frac{\alpha}{b}]+\alpha at}{1+\beta e^{-bt}}$ $a(t) = a(1 + \alpha t)$ $b(t) = \frac{b}{1+\beta e^{-bt}}$	Assume introduction rate is a linear function of testing time, and the fault detection rate function is nondecreasing with an inflexion S-shaped model
P-Z [Pham and Zhang (1997)]	S-shaped and concave	$m(t) = \frac{1}{(1+\beta e^{-bt})} \times [(c + a)(1 - e^{-bt}) - \frac{a}{b-\alpha}(e^{-\alpha t} - e^{-bt})]$ $a(t) = c + a(1 - e^{-\alpha t})$ $b(t) = \frac{b}{1+\beta e^{-bt}}$	Assume introduction rate is exponential function of the testing time, and the fault detection rate is nondecreasing with an inflexion S-shaped model
New Model	S-shaped and concave	$m(t) = a - ae^{-bt} \times (1 + (b + d)t + bdt^2)$ $a(t) = a$ $g(t) = \frac{b^2 t}{1+bt} - \frac{d}{1+dt}$	Assume a constant initial fault content function, and the imperfect fault detection rate combining the fault introduction phenomenon

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Substitute Eq. (7) into Eq. (3) with $m(0) = 0$, we obtain:

$$m(t) = a - ae^{-bt}[1 + (b + d)t + bdt^2] \tag{10}$$

Note that both imperfect debugging and learning curve of the fault detection and introduction are integrated into the proposed model.

Table 1 summarizes the features of the proposed and several existing NHPP models, where in Table 1 the function $a(t)$ is defined as the fault content function, and $b(t)$ is the fault detection function, and $g(t)$ is the imperfect fault detection rate.

2.2.2. *Model Comparison Criterion*

The mean squared errors (MSE) is usually used as a criterion for comparison of goodness-of-fit and predictive power. MSE can be calculated as follows:

$$\text{MSE} = \frac{\sum_{k=1}^{N}[\hat{m}(t_k) - y_k]^2}{N - n} \tag{11}$$

where

y_k: observed number of failures by time t_k
$\hat{m}(t_k)$: estimated number of failures by time t_k
k: testing interval index
N: number of the observations
n: number of parameters in the model

MSE value indicates the overall relative "goodness" of a model after compensation for its complexity in terms of the number of parameters it has. The small the MSE value, the better the model predicts.

3. Model Evaluation and Applications

3.1. *Goodness-of-fit Test: Data from IBM Entry Software Package*

In this section, we examine both the goodness-of-fit of the proposed model and the existing NHPP models using the software failure data collected from testing an online data entry software package at IBM [Ohba (1984a)] (see Table 2). The failures are recorded in days.

We use subset of the above data to fit the models and estimate the parameters. Then we use the remaining ones to compare the predictive power of these models. For illustration purpose, we assume that the software has been tested for 17 days and the software failures during these 17 days are recorded. We estimate the parameters and determine the software reliability models using the first 17 data points and listed the parameter estimate in Table 3. We then utilized all the 21 data to estimate the model parameters, and the last column of Table 3 lists the estimation using all the 21 data.

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Table 2. Failure data from IBM entry software package.

Testing Time (Day)	Failures	Cumulative Failures
1	2	2
2	1	3
3	1	4
4	1	5
5	2	7
6	2	9
7	2	11
8	1	12
9	7	19
10	3	21
11	1	22
12	2	24
13	2	26
14	4	30
15	1	31
16	6	37
17	1	38
18	3	41
19	1	42
20	3	45
21	1	46

The results in Table 3 show that the new model provides consistent estimates for the number of initial faults ($\hat{a}$). That is, $\hat{a}$ estimated using 17 data points are very close to the one estimated using all the 21 data points. This consistency will provide software developers with the precise information of how many initial faults in the software and number of faults remaining in the code at any time in the testing process. The estimates given by other models change when more failure observations are included into the model fitting.

Table 4 summarizes the MSE value for model comparison. MSE is calculated based on 21-day failure data. From the results, we can draw a conclusion that the new model provides the best fit.

3.2. *Evaluation for the Predictive Power: Data from a Real Time Control System*

In this section, the predictive power of the proposed model is evaluated by using a data set collected from testing a program for monitor and real-time control systems [Tohma *et al.* (1991)]. The software consists of about 200 modules and each module has, on average, 1000 lines of a high-level language like FORTRAN. Table 5 records the software failures detected during 111-day testing period. This actual data set is concave overall with two clusters of significant increasing detected faults.

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Table 3. MLEs of model parameters — IBM data.

Model Name	MVF ($m(t)$)	MLEs (17 Data Points)	MLEs (21 Data Points)
Goel-Okumoto (G-O)	$m(t) = a(1 - e^{-bt})$ $a(t) = a$ $b(t) = b$	$\hat{a} = 37435.36$ $\hat{b} = 5.97 \times 10^{-5}$	$\hat{a} = 56051$ $\hat{b} = 3.9 \times 10^{-5}$
Delayed S-shaped	$m(t) = a(1 - (1 + bt)e^{-bt})$ $a(t) = a$ $b(t) = \frac{b^2 t}{1+bt}$	$\hat{a} = 75.36$ $\hat{b} = 0.1995$	$\hat{a} = 71.73$ $\hat{b} = 0.10397$
Inflexion S-shaped	$m(t) = \frac{a(1-e^{-bt})}{1+\beta e^{-bt}}$ $a(t) = a$ $b(t) = \frac{b}{1+\beta e^{-bt}}$	$\hat{a} = 63.25$ $\hat{b} = 0.1558$ $\hat{\beta} = 7.743$	$\hat{a} = 57.37$ $\hat{b} = 0.175$ $\hat{\beta} = 8.5136$
Yamada Exponential	$m(t) = a(1 - e^{-r\alpha(1-e^{(-\beta t)})})$ $a(t) = a$ $b(t) = r\alpha\beta e^{-\beta t}$	$\hat{a} = 17081$ $\hat{\alpha} = 0.725$ $\hat{\beta} = 0.000181$	$\hat{a} = 17264.83$ $\hat{\alpha} = 0.734$ $\hat{\beta} = 0.000173$
Yamada Rayleigh	$m(t) = a(1 - e^{-r\alpha(1-e^{(-\beta t^2/2)})})$ $a(t) = a$ $b(t) = r\alpha\beta e^{-\beta t^2/2}$	$\hat{a} = 645.57$ $\hat{\alpha} = 0.0837$ $\hat{\beta} = 0.0089$	$\hat{a} = 664.19$ $\hat{\alpha} = 0.086$ $\hat{\beta} = 0.0083$
Yamada Imperfect Debugging (1)	$m(t) = \frac{ab}{\alpha+b}(e^{\alpha t} - e^{-bt})$ $a(t) = ae^{\alpha t}$ $b(t) = b$	$\hat{a} = 91920.94$ $\hat{\alpha} = 0.0454$ $\hat{b} = 1.61 \times 10^{-5}$	$\hat{a} = 39252.59.$ $\hat{\alpha} = 0.0185$ $\hat{b} = 4.57 \times 10^{-5}$
Yamada Imperfect Debugging (2)	$m(t) = a[1 - e^{-bt}]\left[1 - \frac{\alpha}{b}\right] + \alpha at$ $a(t) = a(1 + \alpha t)$ $b(t) = b$	$\hat{a} = 1787056$ $\hat{\alpha} = 0.08104$ $\hat{b} = 7.41 \times 10^{-6}$	$\hat{a} = 1671644$ $\hat{\alpha} = 0.02932$ $\hat{b} = 1.0 \times 10^{-6}$
PNZ	$m(t) = \frac{a[1-e^{-bt}][1-\frac{\alpha}{b}]+\alpha at}{1+\beta e^{-bt}}$ $a(t) = a(1 + \alpha t)$ $b(t) = \frac{b}{1+\beta e^{-bt}}$	$\hat{a} = 63.19$ $\hat{b} = 0.1559$ $\hat{\alpha} = 4.98 \times 10^{-5}$ $\hat{\beta} = 7.735$	$\hat{a} = 57.30$ $\hat{b} = 0.175$ $\hat{\alpha} = 4.96 \times 10^{-5}$ $\hat{\beta} = 8.506$
P-Z	$m(t) = \frac{1}{(1+\beta e^{-bt})}[(c + a)(1 - e^{-bt}) - \frac{ab}{b-\alpha}(e^{-\alpha t} - e^{-bt})]$ $a(t) = c + a(1 - e^{-\alpha t})$ $b(t) = \frac{b}{1+\beta e^{-bt}}$	$\hat{a} = 303.71$ $\hat{b} = 0.01125$ $\hat{\alpha} = 0.0625$ $\hat{\beta} = 0.03105$ $\hat{c} = 104.0103$	$\hat{a} = 167.9$ $\hat{b} = 0.0112$ $\hat{\alpha} = 0.1592$ $\hat{\beta} = 0.01785$ $\hat{c} = 100.615$
New Model	$m(t) = a - ae^{-bt}(1 + (b + d)t + bdt^2)$ $a(t) = a$ $g(t) = \frac{b^2 t}{1+bt} - \frac{d}{1+dt}$	$\hat{a} = 60.32$ $\hat{b} = 0.138$ $\hat{d} = 0.011$	$\hat{a} = 60$ $\hat{b} = 0.137$ $\hat{d} = 0.013$

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Table 4. Model comparison — IBM data.

Model Name	MVF ($m(t)$)	MSE (Fit)
Goel-Okumoto (G-O)	$m(t) = a(1 - e^{-bt})$ $a(t) = a$ $b(t) = b$	117.24
Delayed S-shaped	$m(t) = a(1 - (1 + bt)e^{-bt})$ $a(t) = a$ $b(t) = \frac{b^2 t}{1+bt}$	38.0
Inflexion S-shaped	$m(t) = \frac{a(1-e^{-bt})}{1+\beta e^{-bt}}$ $a(t) = a$ $b(t) = \frac{b}{1+\beta e^{-bt}}$	52.01
Yamada Exponential	$m(t) = a(1 - e^{-r\alpha(1-e^{(-\beta t)})})$ $a(t) = a$ $b(t) = r\alpha\beta e^{-\beta t}$	124.81
Yamada Rayleigh	$m(t) = a(1 - e^{-r\alpha(1-e^{(-\beta t^2/2)})})$ $a(t) = a$ $b(t) = r\alpha\beta t e^{-\beta t^2/2}$	36.89
Yamada Imperfect Debugging (1)	$m(t) = \frac{ab}{\alpha+b}(e^{\alpha t} - e^{-bt})$ $a(t) = ae^{\alpha t}$ $b(t) = b$	53.09
Yamada Imperfect Debugging (2)	$m(t) = a[1 - e^{-bt}]\left[1 - \frac{\alpha}{b}\right] + \alpha a t$ $a(t) = a(1 + \alpha t)$ $b(t) = b$	43.87
PNZ	$m(t) = \frac{a[1-e^{-bt}]\left[1-\frac{\alpha}{b}\right]+\alpha a t}{1+\beta e^{-bt}}$ $a(t) = a(1 + \alpha t)$ $b(t) = \frac{b}{1+\beta e^{-bt}}$	34.05
P-Z	$m(t) = \frac{1}{(1+\beta e^{-bt})}[(c + a)(1 - e^{-bt}) - \frac{ab}{b-\alpha}(e^{-\alpha t} - e^{-bt})]$ $a(t) = c + a(1 - e^{-\alpha t})$ $b(t) = \frac{b}{1+\beta e^{-bt}}$	37.13
New Model	$m(t) = a - ae^{-bt}(1 + (b + d)t + bdt^2)$ $a(t) = a$ $b(t) = \frac{b^2 t}{1+bt} - \frac{d}{1+dt}$	32.20

Since the software turns stable after 61 days of testing, we will estimate the model parameters using the first 61 data points and compare them with parameter estimates using all the 111 data points. The results are summarized in Table 6.

There are totally 481 faults detected by the end of 111-day of testing period. As seen from Table 6, the number of initial faults, a, estimated by the new model using the first 61 data points is 482.5, whereas when using all the 111 data is 482.

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Table 5. Failure per day and cumulative failure (those marked with * are interpolated data).

Days	Faults	Cumulative Faults	Days	Faults	Cumulative Faults	Days	Faults	Cumulative Faults	Days	Faults	Cumulative Faults
1	5*	5*	29	2	254	57	2	448	85	0	473
2	5*	10*	30	5	259	58	3	451	86	0	473
3	5*	15*	31	4	263	59	2	453	87	2	475
4	5*	20*	32	1	264	60	7	460	88	0	475
5	6*	26*	33	4	268	61	3	463	89	0	475
6	8	34	34	3	271	62	0	463	90	0	475
7	2	36	35	6	277	63	1	464	91	0	475
8	7	43	36	13	293	64	0	464	92	0	475
9	4	47	37	19	309	65	1	465	93	0	475
10	2	49	38	15	324	66	0	465	94	0	475
11	31	80	39	7	331	67	0	465	95	0	475
12	4	84	40	15	346	68	1	466	96	1	476
13	24	108	41	21	367	69	1	467	97	0	476
14	49	157	42	8	375	70	0	467	98	0	476
15	14	171	43	6	381	71	0	467	99	0	476
16	12	183	44	20	401	72	1	468	100	1	477
17	8	191	45	10	411	73	1	469	101	0	477
18	9	200	46	3	414	74	0	469	102	0	477
19	4	204	47	3	417	75	0	469	103	1	478
20	7	211	48	8	425	76	0	469	104	0	478
21	6	217	49	5	430	77	1	470	105	0	478
22	9	226	50	1	431	78	2	472	106	1	479
23	4	230	51	2	433	79	0	472	107	0	479
24	4	234	52	2	435	80	1	473	108	0	479
25	2	236	53	2	437	81	0	473	109	1	480
26	4	240	54	7	444	82	0	473	110	0	480
27	3	243	55	2	446	83	0	473	111	1	481
28	9	252	56	0	446	84	0	473			

We can draw the following conclusions: (1) the estimates given by both the subset data and the full data set are very close to the actual number of total faults; (2) the estimation is stable and consistent. This indicates that the new model can provide developers with precise information about the total number of initial faults and number of remaining faults at any time during testing phase. Table 7 summaries the MSE values of model prediction.

From Table 7, the proposed model fits and predicts significantly better than other existing models on this data set. This can be explained by the following two reasons: (1) the model incorporates both the testing coverage and imperfect debugging issues and improves the accuracy, even more important, (2) there are some changes of testing efforts during the debugging process in terms of increasing

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Table 6. MLEs of model parameters — the control system data.

Model Name	MVF ($m(t)$)	MLEs (61 Data Points)	MLEs (111 Data Points)
Goel-Okumoto	$m(t) = a(1 - e^{-bt})$	$\hat{a} = 852.97$	$\hat{a} = 497.282$
(G-O)	$a(t) = a$	$\hat{b} = 0.01283$	$\hat{b} = 0.0308$
	$b(t) = b$		
Delayed S-shaped	$m(t) = a(1 - (1 + bt)e^{-bt})$	$\hat{a} = 522.49$	$\hat{a} = 483.039$
	$a(t) = a$	$\hat{b} = 0.06108$	$\hat{b} = 0.06866$
	$b(t) = \frac{b^2 t}{1+bt}$		
Inflexion S-shaped	$m(t) = \frac{a(1-e^{-bt})}{1+\beta e^{-bt}}$	$\hat{a} = 852.45$	$\hat{a} = 482.017$
	$a(t) = a$	$\hat{b} = 0.01285$	$\hat{b} = 0.07025$
	$b(t) = \frac{b}{1+\beta e^{-bt}}$	$\hat{\beta} = 0.001$	$\hat{\beta} = 4.15218$
Yamada	$m(t) = a(1 - e^{-r\alpha(1-e^{(-\beta t)})})$	$\hat{a} = 9219.7$	$\hat{a} = 67958.8$
Exponential	$a(t) = a$	$\hat{\alpha} = 0.09995$	$\hat{\alpha} = 0.00732$
	$b(t) = r\alpha\beta e^{-\beta t}$	$\hat{\beta} = 0.01187$	$\hat{\beta} = 0.03072$
Yamada Rayleigh	$m(t) = a(1 - e^{-r\alpha(1-e^{(-\beta t^2/2)})})$	$\hat{a} = 611.70$	$\hat{a} = 500.146$
	$a(t) = a$	$\hat{\alpha} = 1.637$	$\hat{\alpha} = 3.31944$
	$b(t) = r\alpha\beta t e^{-\beta t^2/2}$	$\hat{\beta} = 0.00107$	$\hat{\beta} = 0.00066$
Yamada Imperfect	$m(t) = \frac{ab}{\alpha+b}(e^{\alpha t} - e^{-bt})$	$\hat{a} = 1795.7$	$\hat{a} = 654.963$
Debugging	$a(t) = ae^{\alpha t}$	$\hat{b} = 0.00614$	$\hat{b} = 0.02059$
Model (1)	$b(t) = b$	$\hat{\alpha} = 0.002$	$\hat{\alpha} = 0.0027$
Yamada Imperfect	$m(t) = a[1 - e^{-bt}]\left[1 - \frac{\alpha}{b}\right] + \alpha at$	$\hat{a} = 16307$	$\hat{a} = 591.804$
Debugging	$a(t) = a(1 + \alpha t)$	$\hat{b} = 0.0068$	$\hat{b} = 0.02423$
Model (2)	$b(t) = b$	$\hat{\alpha} = 0.009817$	$\hat{\alpha} = 0.0019$
PNZ Model	$m(t) = \frac{a[1-e^{-bt}][1-\frac{\alpha}{b}]+\alpha at}{1+\beta e^{-bt}}$	$\hat{a} = 470.759$	$\hat{a} = 470.759$
	$a(t) = a(1 + \alpha t)$	$\hat{b} = 0.07497$	$\hat{b} = 0.07497$
	$b(t) = \frac{b}{1+\beta e^{-bt}}$	$\hat{\alpha} = 0.00024$	$\hat{\alpha} = 0.00024$
		$\hat{\beta} = 4.69321$	$\hat{\beta} = 4.69321$
P-Z Model	$m(t) = \frac{1}{(1+\beta e^{-bt})}[(c + a)(1 - e^{-bt})$	$\hat{a} = 0.920318$	$\hat{a} = 0.46685$
	$- \frac{ab}{b-\alpha}(e^{-\alpha t} - e^{-bt})]$	$\hat{b} = 0.0579$	$\hat{b} = 0.07025$
	$a(t) = c + a(1 - e^{\alpha t})$	$\hat{\alpha} = 2.76 \times 10^{-5}$	$\hat{\alpha} = 1.4 \times 10^{-5}$
	$b(t) = \frac{b}{1+\beta e^{-bt}}$	$\hat{\beta} = 3.152$	$\hat{\beta} = 4.15213$
		$\hat{c} = 520.784$	$\hat{c} = 482.016$
New Model	$m(t) = a - ae^{-bt}(1 + (b + d)t + bdt^2)$	$\hat{a} = 482.5$	$\hat{a} = 482$
	$a(t) = a$	$\hat{b} = 0.0751$	$\hat{b} = 0.081$
	$b(t) = \frac{b^2 t}{1+bt} - \frac{d}{1+dt}$	$\hat{d} = 0.006$	$\hat{d} = 0.007$

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Table 7. Model comparison — the control system data.

Model Name	MVF ($m(t)$)	MSE (Production)
Goel-Okumoto (G-O)	$m(t) = a(1 - e^{-bt})$ $a(t) = a$ $b(t) = b$	11611.42
Delayed S-shaped	$m(t) = a(1 - (1 + bt)e^{-bt})$ $a(t) = a$ $b(t) = \frac{b^2 t}{1+bt}$	935.88
Inflexion S-shaped	$m(t) = \frac{a(1-e^{-bt})}{1+\beta e^{-bt}}$ $a(t) = a$ $b(t) = \frac{b}{1+\beta e^{-bt}}$	590.38
Yamada Exponential	$m(t) = a(1 - e^{-r\alpha(1-e^{(-\beta t)})})$ $a(t) = a$ $b(t) = r\alpha\beta e^{-\beta t}$	12228.25
Yamada Rayleigh	$m(t) = a(1 - e^{-r\alpha(1-e^{(-\beta t^2/2)})})$ $a(t) = a$ $b(t) = r\alpha\beta e^{-\beta t^2/2}$	187.57
Yamada Imperfect Debugging Model (1)	$m(t) = \frac{ab}{\alpha+b}(e^{\alpha t} - e^{-bt})$ $a(t) = ae^{\alpha t}$ $b(t) = b$	8950.54
Yamada Imperfect Debugging Model (2)	$m(t) = a[1 - e^{-bt}]\left[1 - \frac{\alpha}{b}\right] + \alpha at$ $a(t) = a(1 + \alpha t)$ $b(t) = b$	2752.83
PNZ Model	$m(t) = \frac{a[1-e^{-bt}][1-\frac{\alpha}{b}]+\alpha at}{1+\beta e^{-bt}}$ $a(t) = a(1 + \alpha t)$ $b(t) = \frac{b}{1+\beta e^{-bt}}$	2480.7
P-Z Model	$m(t) = \frac{1}{(1+\beta e^{-bt})}[(c + a)(1 - e^{-bt}) - \frac{ab}{b-\alpha}(e^{-\alpha t} - e^{-bt})]$ $a(t) = c + a(1 - e^{\alpha t})$ $b(t) = \frac{b}{1+\beta e^{-bt}}$	102.66
New Model	$m(t) = a - ae^{-bt}(1 + (b + d)t + bdt^2)$ $a(t) = a$ $b(t) = \frac{b^2 t}{1+bt} - \frac{d}{1+dt}$	9.37

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number of testers, the proposed model did a better job in capturing these changes and therefore improved the predictive power.

4. Conclusions

In this study, a NHPP software reliability model that incorporates fault introduction phenomenon and testing coverage information is developed. The proposed model does not require any specific assumptions for the fault content function. Furthermore, this model incorporates fault introduction phenomenon and testing coverage information into error detection. The results show that the number of initial faults in the software estimated by this model is consistent, in other words, it does not change significantly with time. This model has technical merit in estimating the number of initial faults consistently as one progresses through a testing phase. Therefore it can be used to determine the number of remaining faults in the software at any time point during testing process. The proposed model also provides testing coverage measure, which will help developers to evaluate how much of the software has been executed and how much more effort is needed for further improvement.

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